FACE Prep

Operating System - Coding | REC | 3 Feb

Course Progress 11%

Go Back

Back To Practice

Q1

Save

First-Come,First-Served CPU Scheduling Algorithm

Completed

Write a C program to implement the First-Come, First-Served (FCFS) CPU Scheduling Algorithm. The program should take the number of processes, their Arrival Times (AT), and Burst Times (BT) as input. Calculate and display the Completion Time (CT), Turnaround Time (TAT), and Waiting Time (WT) for each process, along with the Average Turnaround Time and Average Waiting Time.

To evaluate the performance of the algorithm, we use the following formulas:

- **Completion Time (CT):** The time at which a process finishes execution.
- **Turnaround Time (TAT):** Total time taken from arrival to completion.
- $TAT - CT - AT$
- **Waiting Time (WT):** The time a process spends waiting in the ready queue.
- $WT - TAT - BT$

Input Format

1. The first input line contains an integer representing the number of processes.
2. The next lines contain two integers for each process, where the first integer represents the Arrival Time (AT) and the second integer represents the Burst Time (BT).

Select Language : C (GCC 9.2.0)

```c
#include <stdio.h>
int main()
{
    int n,i;
    float avg_tat=0,avg_wt=0;
    printf("Enter number of processes: \n");
    scanf("%d",&n);
    int at[n],bt[n],ct[n],tat[n],wt[n];
    for(i=0;i<n;i++)
    {
        printf("Enter Arrival Time and Burst Time for P%d: \n",i+1);
        scanf("%d %d",&at[i],&bt[i]);
    }
    ct[0]=at[0]+bt[0];
    for(i=1;i<n;i++)
    {
        if(ct[i-1] < at[i])
            ct[i]=at[i]+bt[i];
        else
            ct[i]=ct[i-1]+bt[i];
    }
    for(i=0;i<n;i++)
```

Test Cases 2

Run Sample Cases

Run All Cases

Custom Test Case

> Custom Test

Sample Test Case

> Test Case - 1

FACE Prep

Operating System - Coding | REC | 3 Feb

Course Progress 11%

Go Back

Back To Practice

Q1

Save

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```c
    }
    ct[0]=at[0]+bt[0];
    for(i=1;i<n;i++)
    {
        if(ct[i-1] < at[i])
            ct[i]=at[i]+bt[i];
        else
            ct[i]=ct[i-1]+bt[i];
    }
    for(i=0;i<n;i++)
    {
        tat[i]=ct[i]-at[i];
        wt[i]=tat[i]-bt[i];
        avg_tat+=tat[i];
        avg_wt+=wt[i];
    }
    avg_tat/=n;
    avg_wt/=n;
    printf("\nAverage Turnaround Time: %.2f\n",avg_tat);
    printf("Average Waiting Time: %.2f\n",avg_wt);
    return 0;
}
```

Test Cases 2

Run Sample Cases

Run All Cases

Custom Test Case

> Custom Test

Sample Test Case

> Test Case - 1